

EXPERIENCE

- **Pramatra Space** Bengaluru, India
Research Engineer July 2024 - Present
 - **Digital Design:** Lead the development of electronic systems for operations of the quantum key distribution (QKD) payload onboard a CubeSat as well as for a ground-based use case. This includes the simulation and implementation of post-processing algorithms for several QKD Key Reconciliation Protocols on Xilinx FPGAs and SoCs.
 - **Circuit Design:** Completed the end-to-end circuit design for a Xilinx SoC-based space payload used for carrying out quantum communication experiments in low earth orbit.
 - **System Design:** Designed the mixed-signal electronic control architecture for the control of the Photonics components used for entangled photon generation and detection.
- **Indian Institute of Science (IISc Bangalore)** Bengaluru, India
Project Associate Dec 2023 - July 2024
 - **Contributions:** Worked on the implementation of Signal Processing schemes on RFSoc-based hardware for Quantum circuits application. Part of the team at Quantum Technologies Lab under the supervision of Dr. Baladitya Suri.
 - **Design of readout system for quantum circuits and correlation (g2) measurement using hardware acceleration:** Designed a readout scheme for measuring the purity of a microwave single-photon source and characterizing the output of a Josephson Parametric Converter (JPC). This involved the generation of excitation signal for the photon source and characterizing the output of the signal by elimination of uncorrelated noise and isolation of extremely weak signals produced by the received photons.
- **Indian Space Research Organisation (SAC, ISRO)** Ahmedabad, India
Graduate Apprentice 2022 - Dec 2023
 - **Contributions:** Worked on Research Projects in Digital Design and Signal Processing domains with applications in Aerospace and Applied Physics. Collaborating with a team of competent engineers for the end-to-end development of technologies for Space Radar Applications.
 - **FPGA-based TDC Implementation & Correlation measurement for Entangled Photon Quantum Radar:** Developed an FPGA-based dual-channel Time-to-Digital Converter module for the Time-of-flight calculation of entangled photons with a precision of 20 ps. This was a 15X more precise implementation as compared to the legacy designs. The system was implemented in a Kintex 7 FPGA on an NI PXI FPGA card.
 - **High-configurability Stepper Motor Control Topology for Space Applications:** Designed and implemented a novel Stepper Motor control topology with multi-parameter control inside Virtex-5 FPGA.

EDUCATION

- **Pune Institute of Computer Technology (PICT)** Pune, India
Bachelor of Engineering (B.E.) in Electronics and Telecommunication Engineering 2018 - 2022
 - **CGPA:** 9.45/10.0

PUBLICATIONS

- R. Dhingani, M. Das, **C. Alpe**, H. Patel and B. S. Kumar, “FPGA based Variable Scan Rate Generic Stepper Motor Drive Algorithm for mm-Wave Sounder Payloads”, 2024 IEEE Space, Aerospace and Defence Conference (SPACE), Bangalore, India, 2024, pp. 203-206, doi: 10.1109/SPACE63117.2024.10668030.

SKILLS

- **Digital VLSI:** FPGA Workflows, System-on-Chip (SoC) workflows, HDL (VHDL/Verilog), Verification (Questa/Modelsim), CDC, STA, Strong Digital logic principles
- **Circuit Design:** Circuit Design, SPICE, Hardware Design in KiCad, Autodesk Eagle and Zuken Cadstar
- **System-level:** System Modelling with MATLAB, Simulink, and Labview (Including NI FPGA Design)
- **Programming:** Python, C, C++, Data Structures, Object-oriented programming, Robot Operating System
- **Computer Engineering:** Embedded Programming, Microcontroller architecture

OTHER PROJECTS

- **Implementation of high-speed CORDIC Algorithm for RFSoc-based Quantum Readout(Sep. 2021 - March 2022):** This project included the design and development of a high-speed and low-resource CORDIC algorithm for use in a Quantum Computing Readout System. The resulting design performed at-par with most IP-ready CORDIC algorithms and was tailored to provide significantly better clock rate and configurability.
- **Autonomous Control of Greenhouse Robot for Tomato Picking in ROS and Gazebo (Sep. 2021 - March 2022):** It involved the development of algorithms for autonomous navigation of an Agri-Robot in a greenhouse environment and controlling a Robotic arm for picking ripe tomatoes using Computer Vision. The simulation was done in the Gazebo Simulation Environment and the entire project was built using the ROS framework.
- **Solar-Powered Reconnaissance Robot (Aug. 2021 - May 2022):** The project involved the development of a solar-powered robot that can infiltrate and monitor critical defensive sites and perform real-time surveillance for extended periods using Night Vision capabilities with stealthy operation and a controlling station for control and monitoring of the robot.

VOLUNTEERING

- **Yuva Bengaluru Trust** Bengaluru, India
Teaching Volunteer *May 2024 - Present*
 - **Contributions:** Teaching the English language and other essential life skills to school children from underprivileged government schools every weekend. This also involves participating in different charitable activities such as food distribution and career counseling for high school students.
- **PICT Robotics Club** Pune, India
Sensing Lead and Technical Team Member *Sept 2018 - July 2020*
 - **Contributions:** Led the team in the design and implementation of optimal sensing strategies. Contributed to the Electronics as well as the Coding Department at the Club. Was also involved in the Electronics Interface of different subsystems of Robots for DD Robocon 2019-20, conducted at IIT Delhi.

ACHIEVEMENTS

- **e-Yantra Robotics Competition:** Team achieved a rank of 14 (Top 1 percentile) at the e-Yantra Robotics Competition, IIT Bombay which involved the design and simulation of robotics software for an Agriculture Robot.
- **PICT Cretronix 2019:** Achieved 1st rank at Cretronix Circuit Design event in 2019. This technical event was organized as a part of PICT IEEE's technical fest 'Credenz'.
- **PICT Cretronix 2020:** Achieved 2nd rank at Cretronix Circuit Design event in 2020. This technical event was organized as a part of PICT IEEE's technical fest 'Credenz'.

SOFT SKILLS AND EXTRA CURRICULARS

- Represented the institutes at Undergraduate and High School Level Table Tennis competitions with Silver Medal in Inter-College Tournament
- Avid bird-watcher, photographer and sketch artist